

Experiment Report

GROUP MEMBERS:

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DEVICE USED: iPhone 15 Pro

RESOLUTION OF DEVICE:

Main Camera:

- 48 MP wide (24mm, f/1.78, sensor-shift OIS)
- 12 MP ultra-wide (13mm, f/2.2)
- 12 MP telephoto (77mm, f/2.8, 3x optical zoom)

Photo Modes: Normal resolution (12 MP) and High-Resolution (24 MP / 48 MP ProRAW)

Video Capability: Up to 4K @ 60 fps

Other Features: Night Mode, Deep Fusion, Smart HDR, Motion Stabilization

Condition A – Bright Sunlight

Photo 1: Front view, strong sunlight, shadow visible



Photo 2: Side angle, bright sunlight, sharp colors



Photo 3: Slightly angled, strong sunlight



Condition B – Shadow:

The box appears dimmer, and fine details like the zigzag pattern are less prominent.

Photo 1: (taken under a scooter shadow)



photo 2:(taken under a banner shadow)



Photo 3:(taken under a scooter shadow)



Condition C – Dim Lighting

Photo 1:



Photo 2:



Photo 3:



Resolution (Normal vs. High):

High Resolution:



Normal Resolution:



High-resolution images preserved tiny details such as the crown symbols. Normal resolution lost some sharpness, especially when zoomed in.

Motion (Still vs. Moving):

Still:



Moving:



Still photos were crisp and easy to recognize.

Motion blur caused distortion: the unicorn's outline looked stretched.

Observations:

Lighting Effect:

Lighting was the most critical factor. Bright sunlight gave the best clarity and color. In shadow and dim light, the pink shade became dull, and details blurred into the background.

Resolution Effect:

High-resolution made a clear difference when analyzing small designs. The crowns and outlines looked sharp, while in normal resolution they appeared fuzzy.

Motion Effect:

Still images were always better. Moving captures showed streaks and blur, reducing both clarity and recognizability.